

Harsh Kumar

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EDUCATION

Computer Science (B.C.Sc.) Honours Co-op, Mathematics Minor

Jan. 2024 – Present

University of Manitoba

Winnipeg, MB

- **Awards & Honours:** Dean's Honour List 2024, 2025 (Winter, Summer).
GPA: 3.74/4.5

SKILLS

Languages: Java, JavaScript, TypeScript, C++, C, Python, HTML/CSS

Tools & OS: Git and GitHub, VS Code, Vim, Testing & Debugging, Linux and UNIX-based OS, npm/Vite

Frameworks & Database: React, PGLite (PostgreSQL), MongoDB, Node.js, Rasa

Productivity Tools: Spreadsheets, Google Workplace, L^AT_EX, Discord, Microsoft 365 Suite

EXPERIENCE

Student Office Assistant

Sept. 2026 – Present

Student Transitions and Success Centre, University of Manitoba

Winnipeg, MB

- Mentor undergraduate students through university transitions, challenges, and campus resources. While, building supportive relationships through active listening, empathy, and clear communication.
- Identify student needs and connect mentees with appropriate campus resources while maintaining professionalism and confidentiality.
- Facilitate workshops and activities that encourage student engagement, confidence, helping them build a connection with the university community.
- Manage mentoring responsibilities independently while balancing multiple priorities and collaborating with program staff and other mentors.

Liquor Store (MBLL) Courtesy Clerk

May 2024 – Present

River & Osborne Liquor Mart

Winnipeg, MB

- Deliver high-quality customer service to a diverse clientele in a fast-paced retail environment.
- Receive, restock, and organize inventory while maintaining accurate stock levels and store presentation.
- Collaborate with team members and communicate with management to ensuring smooth daily operations.
- Train new employees one-to-one on store policies, procedures, and POS system to optimize productivity.
- Conduct regular safety and wellness checks in partnership with the Osborne Village BIZ, covering customers inside the store and in the surrounding area, resulting in fewer incidents on site.

EXTRACURRICULAR & COMMUNITY INVOLVEMENT

Student Services Programmer (Operations)

June 2026 – Present

Science Students' Association (Faculty of Science), University of Manitoba

Winnipeg, MB

- Develop advertising strategies to increase awareness and sales of science-complex lockers.
- Promote graduate photography sessions through coordinated advertising and photographer liaison.
- Manage SSA rentals and inventory, ensuring efficient distribution, returns, and inventory checks.

Membership Experience Executive

April 2026 – Present

.devClub, University of Manitoba

Winnipeg, MB

- Gather student feedback, suggestions, and responses to improve engagement.
- Communicate with other student organizations to ensure an inclusive and supportive club environment.
- Collaborate with external, promotions, and events teams to enhance member experience.

Research Involvement

Jan. 2026 – Present

CORE AI Lab [✉](#), University of Manitoba

Winnipeg, MB

- Attend weekly AI lab meetings discussing topics such as lossy data compression and rate-distortion theory.
- Collaborated with graduate researchers on AI/ML concepts and research topics.
- Engage with faculty, including *Dr. Sadaf Salehkalaibar*, on advanced machine learning topics.

MGEU Union Member

Manitoba Government and General Employees' Union 

May 2024 – Present

Winnipeg, MB

- Attend bargaining meetings and observed real-world negotiation processes.
- Built communication and collaboration skills in a professional setting.
- Voice workplace concerns through union discussions, strengthening advocacy and collaboration skills.

Science Rendezvous Volunteer

Faculty of Science, University of Manitoba

May 2024, May 2025

Winnipeg, MB

- **2025** : Facilitated interactive STEM activities as a booth volunteer for Engineering for Kids, engaging youth and promoting interest in engineering concepts.
- **2024** : Assisted as a show usher for the Manitoba Prairie Wildlife & Rehabilitation, directing big school groups to help conduct the show smoothly. Demonstrating strong communication, organization, and teamwork skills.

PROJECTS

Smart Hospital Management System | MongoDB, Express.js, React (Vite), Node.js

Personal

- Built a full-stack MERN application with MVC architecture supporting authentication, OTP verification, and role-based admin panel (CRUD).
- Implemented patient management, scheduling, medical orders, and search/filter functionality.
- Integrated real-time chat (Socket.io), notifications, file uploads, and geolocation maps (Leaflet/OpenStreetMap).
- Designed scalable backend with REST APIs and seeded database for testing and development.

Bison Bot | Angular, Rasa, Python, Vector Search, Ollama, CSS, FIASS, TypeScript, HTML

Hackathon

- Developed an AI-powered academic advisor chatbot for UofM students using natural language queries.
- Led backend development in a team of five, trained Rasa NLU model, handled custom actions, and structured domain-specific knowledge base.
- Integrated vector search with local LLMs (Ollama) to enhance contextual and offline responses.
- Connected backend to Angular front-end via REST APIs for seamless deployment.

Reverse Fortune Teller | HTML, CSS, JavaScript, Python

Hackathon

- Collaborated in a team of four to develop a reverse fortune teller web application within a limited development time frame.
- Developed a chatbot that generates predictions about users' past experiences based on personality traits and user-provided inputs.
- Participated in project planning, feature implementation, and iterative testing to deliver a functional prototype by the end of the event.

Candy Clicker | HTML, CSS, TypeScript, PGLite, Vite

Academic

- Built a Cookie Clicker style game using MVC architecture, SOLID principles, and Design by Contract, producing maintainable and scalable code.
- Designed persistent state management using PGLite to store player progress, upgrades, and autobuy configurations.
- Developed a probabilistic Markov chain autobuy algorithm to optimize automated purchasing decisions. While being mindful about the resource allocation.
- Performed testing through Vitest and ensured appropriate test coverage. Integrated new code while testing, to catch code breakage.

INTERESTS

LeetCode (50+ Problems) :

- Practice topics including arrays, backtracking, math, recursion, and dynamic programming etc.

Running :

- Trained for, and completed the 10K at Manitoba Marathon 2026.

TRAININGS COMPLETED

🔧 Respectful Workplace, WHIMIS, Fire Safety, Anti-Racism, Hard Target (Robbery Prevention), BASIC.